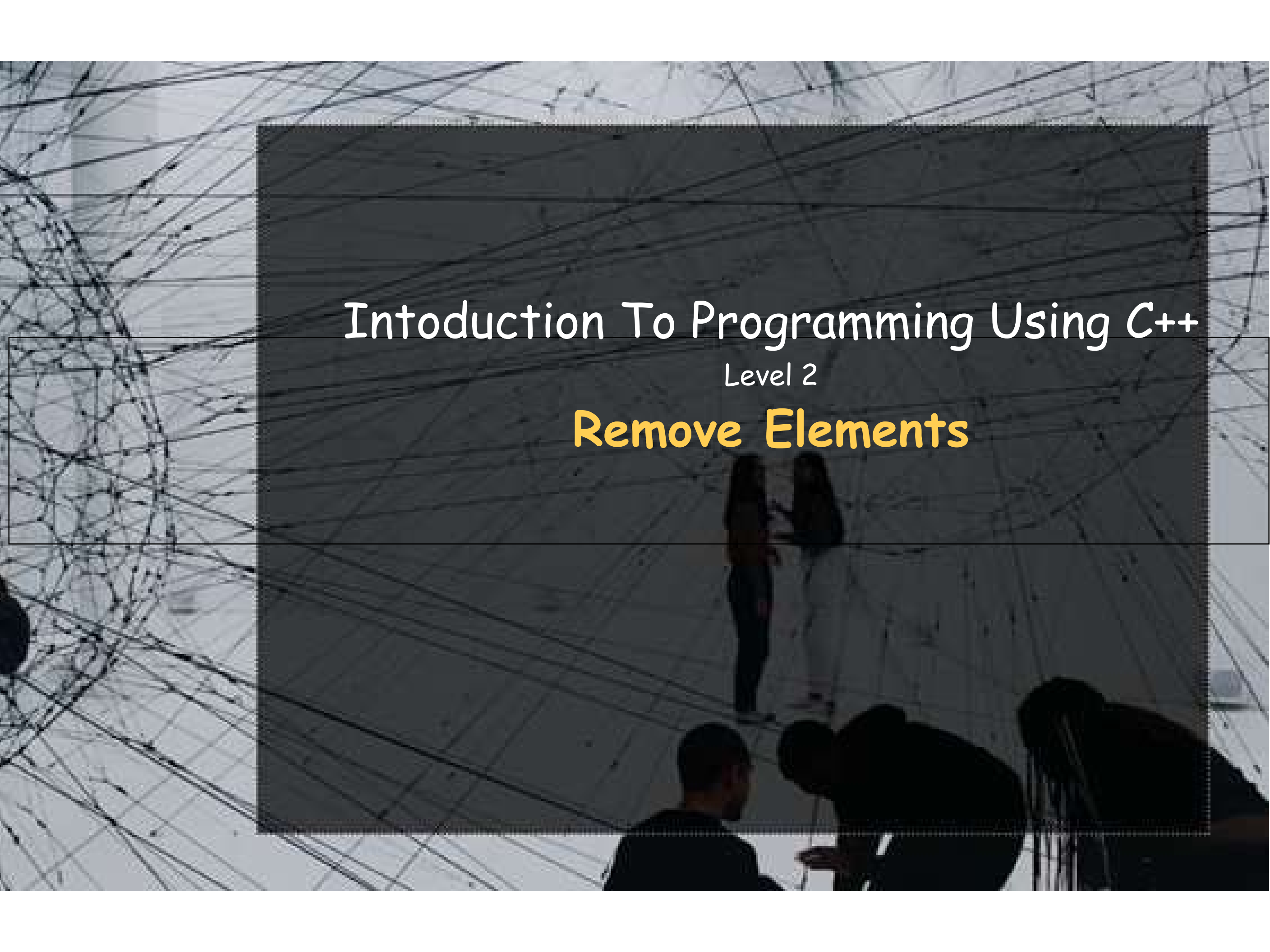




Introduction To Programming Using C++ Level 2

Remove Elements



Introduction To Programming Using C++

Level 2

Remove Elements

Definition

- ◆ `pop_back()` is a member function of `std::vector` used to remove the last element from a Vector.
- ◆ It does not take any parameters because it always removes the last element.
- ◆ When elements are added using `push_back()` and removed using `pop_back()`, the operation follows the LIFO principle.
- ◆ LIFO = Last In, First Out.

Explanation

`pop\_back()` Function

Syntax:

`vNumbers.pop_back();`

 It removes the last element of the Vector.

 It reduces the Vector's `size()` by one.

 It does not return the removed element.

💡 Example

```
int main()
{
    vector<int> vNumbers;
    vNumbers.push_back(10);
    vNumbers.push_back(20);
    vNumbers.push_back(30);
    vNumbers.push_back(40);
    vNumbers.push_back(50);
    vNumbers.pop_back();
    vNumbers.pop_back();
    cout << "Numbers of Vector:\n\n";
    for (int Number : vNumbers)
    {
        cout << Number << endl;
    }
    return 0;
}
```

➡ Result

```
Numbers of Vector:
10
20
30
```

Note

- ◆ To find the number of elements in a Vector, use `size()` :
`cout << "Vector Size = " << vNumbers.size() << endl;`

⊘ Warning

- ◆ Do not call `pop_back()` when the Vector is empty.
- ◆ Always make sure the Vector contains at least one element before calling `pop_back()`:

```
if (!vNumbers.empty())  
{  
    vNumbers.pop_back();  
}
```
- ◆ After removing elements with `pop_back()` or `clear()`, do not access elements that no longer exist.
- ◆ The `clear()` function removes all elements, so the Vector's `size()` becomes `0`.



Comparison

▨ `pop\_back()`

- ✓ Removes only the last element.
- ✓ Reduces the Vector's `size()` by one.
- ✓ Takes no parameters.
- ✓ Example: `vNumbers.pop\_back();`

▤ `clear()`

- ✓ Removes all elements from the Vector.
- ✓ Makes the Vector empty.
- ✓ Example: `vNumbers.clear();`

▥ `erase()`

- ✓ Removes an element or a range of elements at a specified position.
- ✓ Requires an iterator or iterator range.

✦✦ Summary

- ✓ `push_back()` adds an element to the end of a Vector.
- ✓ `pop_back()` removes the last element.
- ✓ `pop_back()` does not take any parameters.
- ✓ `empty()` checks whether the Vector contains no elements.
- ✓ `size()` returns the number of elements in the Vector.
- ✓ `clear()` removes all elements at once.
- ✓ `erase()` can remove elements from a specific position or range.
- ✓ `push_back()` + `pop_back()` can be used to implement LIFO behavior.
- ✓ Always check that the Vector is not empty before calling `pop_back()`.

Thank you 😊